The Hubble Constant in the T0 Model: Energy Loss Through the Universal ξ-Field

Contents

.1 Introduction: Rethinking the Hubble Constant 1
.2 The Universal ξ-Field 2
.3 Energy Loss Mechanism 2
Fundamental Energy Loss Equation 2
Solution for Cosmological Distances 2
.4 Connection to the Hubble Constant 3
.5 Dimensional Analysis 3
.6 Theoretical Advantages 3
.7 Conclusion 4

Abstract

The T0 model interprets cosmological redshift not as expansion but as energy loss of photons in the universal ξ-field. This document treats the field-theoretic energy loss mechanism and its connection to the Hubble constant. The quantitative derivation of $H_0 \approx 66.2$ km/s/Mpc, unit conversion, frequency independence, and the resolution of the Hubble tension are centralized in Document 026 (Geometric Cosmology).

.1 Introduction: Rethinking the Hubble Constant

The conventional interpretation of Hubble's law assumes galaxies recede due to expanding space ($v = H_0 d$). This expansion paradigm requires 69% dark energy, suffers from persistent measurement tensions, and fine-tuning problems.

The T0 model offers a fundamentally different perspective: the universe is static, and the observed redshift arises from energy loss of photons as they propagate through the universal ξ-field.

The Hubble constant thus transforms from a measure of space expansion into a characteristic energy loss rate.

The time-energy duality ($\Delta E \cdot \Delta t \geq \hbar/2$) forbids a temporal beginning of the universe: a finite time interval from a singularity would require infinite energy uncertainty. The universe must have existed eternally, making space expansion unnecessary for explaining cosmic observations.

.2 The Universal ξ-Field

The cornerstone of the T0 model is the universal geometric constant:

$$\xi = \frac{4}{3} \times 10^{-4} = 1.3333\ldots \times 10^{-4} \tag{1}$$

This dimensionless constant determines a characteristic energy scale:

$$E_\xi = \frac{1}{\xi} = 7500 \quad \text{(natural units)} \tag{2}$$

The ξ-field permeates all of space and mediates interactions between photons and the vacuum. Its dynamics are described by the wave equation:

$$\Box E_{\text{field}} = \left(\nabla^2 - \frac{\partial^2}{\partial t^2}\right) E_{\text{field}} = 0 \tag{3}$$

.3 Energy Loss Mechanism

Fundamental Energy Loss Equation

Photons lose energy through interaction with the ξ-field. The rate of energy loss follows the differential equation:

$$\frac{dE}{dx} = -\xi \cdot E \tag{4}$$

The negative sign indicates energy loss; the linear dependence on E ensures that the resulting redshift is frequency-independent (see Document 026 for the geometric justification).

Solution for Cosmological Distances

For cosmological observations ($\xi x \ll 1$), the perturbative solution yields:

$$E(x) = E_0 \left(1 - \xi \cdot x\right) \tag{5}$$

The redshift is therefore:

$$z = \frac{E_0 - E(x)}{E(x)} \approx \xi \cdot x \tag{6}$$

This relationship reproduces the observed linear Hubble law without space expansion.

.4 Connection to the Hubble Constant

Comparison with Hubble's law $z = H_0 d/c$ yields in natural units ($c = 1$):

$$\boxed{H_0^{\text{T0}} = E_H/\hbar, \quad E_H = E_0 \cdot \xi^{41/4} = 1.41 \times 10^{-33}\ \text{eV}} \tag{7}$$

Result: $H_0^{\text{T0}} \approx 66.2$ km/s/Mpc (-1.9% deviation from Planck). The exponent $41/4$ requires physical justification from ξ-field theory.

For the complete numerical derivation, the step-by-step unit conversion (natural units → SI → km/s/Mpc), comparison with experimental measurements, and the resolution of the Hubble tension, see **Document 026** (Geometric Cosmology, Sections 4–6).

.5 Dimensional Analysis

Verification of dimensional consistency in natural units:

Quantity	T0 Expression	Dimension	Status
Geometric constant	$\xi = 4/3 \times 10^{-4}$	$[1]$	✓
Energy scale	$E_\xi = 1/\xi$	$[E]$	✓
Energy loss rate	$dE/dx = -\xi \cdot E$	$[E^2]$	✓
Redshift	$z = \xi \cdot x$	$[1]$	✓
Hubble parameter	$H_0 = E_H/\hbar$	$[T^{-1}]$	✓

.6 Theoretical Advantages

The reinterpretation of the Hubble constant as an energy loss rate solves several problems:

1. **Elimination of dark energy:** The apparent cosmic acceleration arises naturally from distance-dependent energy loss. No exotic energy form with negative pressure is required.
2. **Fine-tuning problems solved:** The horizon problem vanishes since all regions have been in causal contact over infinite time. The flatness problem disappears for lack of initial conditions.
3. **Parameter reduction:** Where ΛCDM requires six independent parameters, T0 derives all cosmological observables from the single geometric constant ξ.

.7 Conclusion

$$\boxed{H_0^{\text{T0}} = E_H/\hbar \approx 66.2\,\text{km/s/Mpc}, \quad E_H = E_0 \cdot \xi^{41/4}} \tag{8}$$

The universe is not expanding. The Hubble constant measures energy loss in the ξ-field, not recession. The detailed derivation via finite element simulation of the T0 vacuum, frequency independence as geometric proof, and the resolution of the Hubble tension are found in Document 026.

Bibliography

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